

MANTHAN PENDKE

Software Development Engineer | Full-Stack & Backend Systems

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SUMMARY

Software developer with hands-on experience designing relational database schemas, building REST APIs, and developing full-stack web applications using Java, JavaScript, and modern frameworks. Strong foundation in object-oriented design, data structures, and system fundamentals, with a track record of shipping authentication, workflow, and dashboard features end to end.

TECHNICAL SKILLS

Languages: Core Java, JavaScript, C++, Python

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, System Design Basics

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Firebase (Auth, Realtime DB, Cloud Functions), Java (OOP-based modules)

Frontend: HTML5, CSS3, JavaScript, React.js

Databases: SQL Server (SSMS), MySQL, MongoDB — ER modeling, normalization

Tools & Cloud: AWS, Git, GitHub, Functional/UI/DB Testing

AI/ML (applied): TensorFlow, PyTorch, OpenCV, RAG, LangChain

EXPERIENCE

Developer Intern — Endava Solutions

Jan 2026 – May 2026

- Designed and implemented a relational database schema in SQL Server (SSMS), applying ER modeling, normalization, and relationship mapping to support a multi-module application.
- Built backend modules in Java using object-oriented programming principles to handle data processing and core business logic.
- Developed responsive web interfaces with HTML, CSS, and JavaScript for authentication, profile management, job posting, and application workflows.
- Performed functional, database, and UI testing, identifying and resolving defects to improve system reliability and overall user experience.

AI Intern — Expleo

July 2025 – Aug 2025

- Built a full-stack predictive maintenance system for CNC machines, integrating ML-driven alarm forecasting into a production workflow.
- Designed and developed a React.js dashboard for real-time alarm visualization, connecting frontend views to backend analytics services.
- Implemented data pipelines (RAG, SMOTE) to feed structured, high-quality input into downstream classification services.
- Collaborated across model and application layers to ship root-cause analysis features using historical sensor and operational data.

Backend Developer Intern — SortUs

May 2025 – Jul 2025

- Developed and maintained backend architecture for a web application using Node.js and Express.js.
- Designed and implemented RESTful APIs for location-based data fetching and core application services.
- Integrated MongoDB for scalable data storage with efficient query handling, and used Firebase for authentication, real-time database, and cloud functions.
- Implemented JWT (JSON Web Token) authentication for secure user sessions and API access control.
- Collaborated with frontend developers using React.js to ensure seamless integration between backend services and UI.
- Optimized backend performance and API response time through efficient query design and asynchronous processing.

PROJECTS

Location-Optimized Web System for Jobs

- Developed an online recruitment platform connecting job seekers and employers, covering the full hiring workflow.
- Implemented secure authentication for user accounts and real-time job tracking with location-based search.
- Designed an intuitive job posting and application system, with backend logic for efficient hiring decisions.
- Integrated location APIs to power personalized, proximity-based job recommendations.

Disaster Information Aggregation Software

- Built an end-to-end data pipeline integrating visual, textual, and geospatial inputs into a real-time alerting system.
- Implemented image preprocessing, OCR, and object detection components within a production-style processing pipeline.
- Co-authored an IEEE research paper on the system's multimodal AI architecture.

College Prediction System

- Built a full prediction workflow — data preprocessing, model training, and serving — for college/branch recommendations.
- Applied MLOps practices for deployment, monitoring, and continuous improvement of the prediction service.

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services (May 2026)
Introduction to Large Language Models (LLMs) — Udemy (April 2025)

EDUCATION

B.Tech, Electronics & Telecommunication Engineering — 8.89 CGPA	Nov 2022 – June 2026
HSC — 74.17%	2021 – 2022

ACHIEVEMENTS

Published an IEEE research paper on Disaster Information Aggregation using Multimodal Artificial Intelligence, integrating NLP, Computer Vision, and Machine Learning techniques.